Operation

During the positive half cycle of secondary voltage, the end A of the secondary winding becomes positive and end B negative. This makes the diode D_1 forward biased and diode D_2 reverse biased. There fore diode D_1 conducts while diode D_2 does not. The conventional current flow is through diode D_1, load resistor RL and the upper half of secondary winding as shown by the dotted arrows.

During the negative half cycle, end A of the secondary winding becomes negative and end B positive. Therefore diode D_2 conducts while diode D_1 does not. The conventional current flow is through diode D_2, load R_L and lower half winding as shown by solid arrows. Referring to fig 10.3 it may be seen that current in the load R_L is in the same direction for both half-cycle of input a.c. voltage. Therefore d.c. is obtained across the load R_L. The peak inverse voltage (piv) of this rectifier is $2V_m$

In recent days center tap full wave rectifier is not used. It is replaced by full wave bridge rectifiers. Because of its application, full wave bridge rectifier is mostly used.

(b) Full wave Bridge Rectifier

Circuit Details

The need for a center tapped power transformer is eliminated in the bridge rectifier. It contains four diodes D_1, D_2, D_3 & D_4 connected to form bridge as shown in fig 10.4. The a.c supply to be rectified is applied to the diagonally opposite ends of the bridge through the transformer. Between other two ends of the bridge, the load resistance R_L is connected.

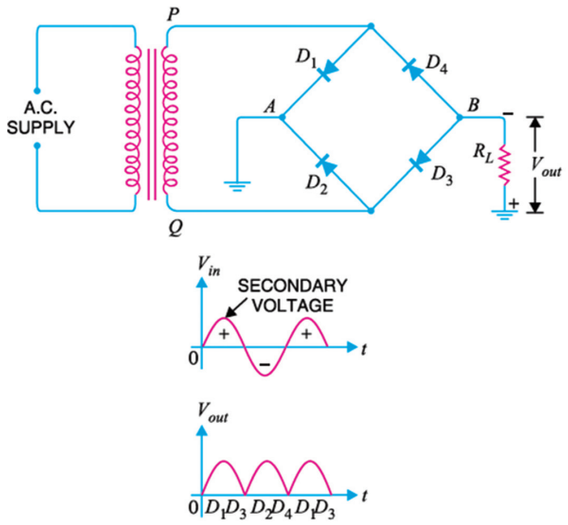


Fig 10.4 *Full Wave Bridge Rectifier-Wave Form*

Operation

During the positive half cycle of secondary voltage, the end P of the secondary winding becomes positive and end 'Q' negative. This makes diode D_1 and D_3 forward biased while diodes D_2 and D_4 are reverse biased. Therefore, only diodes D_1 and D_3 will conduct. These two diodes will be in series through the load R_L as shown in fig 10.5 (i). The conventional current flow is shown by dotted arrows. It may be seen that current flows from A to B through the load R_L

During the negative half-cycle of secondary voltage, end P becomes negative and end Q positive. This makes diodes. D_2 and D_4 forward biased whereas diodes D_1 and D_3 are reverse biased .Therefore, only diodes D_2 and D_4 conduct. These two diodes will be in series through the load R_L as shown in fig 10.5(ii). The current flow is shown by the solid arrows. It may be seen that again current flows from A to

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